

Sensoria's sensors as being skin-safe, having tested these against several pH ranges. These are also less than 1mm in thickness, providing flexibility in form factor for a wide range of clinical applications like diabetic foot management, prosthetics and rehabilitation.

OMSignal's OMbra is also targeted at athletes. Sensors in the sports bra monitor body signals like real-time breathing, heart rate and body movement metrics. This data is sent to an app, which suggests running zones, provides breathing rhythm analysis for increased running efficiency, maximises fat burn, reduces the risk of injury and unnecessary fatigue, and more.

Another much talked about piece of smart clothing is Hexoskin Smart, which has been acclaimed as a biometric shirt. This sports shirt is made of comfortable, durable, machine-washable, odour-resistant Italian fabric, and houses a range of sensors that monitor and record your heart rate, breathing and movement. It works with mobile apps as well as sports watches to provide insights on the intensity of workout, fatigue and recovery, calories burned and quality of sleep.

Hexoskin has 14-plus hours of battery life, and supports Bluetooth connectivity. According to news reports, Hexoskin is also collaborating with Analog Devices and Microsoft to develop an innovative Internet of Things (IoT) solution for team performance management. The solution would allow coaches to monitor players' movements, heart rate and other useful metrics gathered by sensor-equipped vests worn by the players. This information, along with location and environmental data, is sent to a secure, Cloud based analytics platform via Microsoft's Azure IoT technology.

Samsung, too, has a few products up its sleeve. Its workout shirt, Body Compass, monitors biometric data, while its golf shirt includes weather and ultra-violet (UV) rating monitoring.

For the overly fashion conscious,



Fig. 3: Exchange business cards, control devices and do more with Samsung's NFC-enabled smartsuit (Image courtesy: Samsung)

of course, there is Ralph Lauren's PoloTech Shirt, which does pretty much the same stuff like monitoring breathing, heart rate and so on. The PSAL/067 PoloTech Biometric Sensing Technology is screen-printed on the right sleeve, and there is a detachable black-box that uses Bluetooth to transmit key data directly to your iPhone or Apple Watch. This black-box has to be removed before washing. PoloTech shirt is powered by OMSignal.

Flex and MAS Holdings, which have forged a long-term relationship to advance smart clothing, have developed several interesting products. Lumo Run is one such solution developed with Lumo Bodytech. Designed as running shorts and capris, it is supposed to be an athlete's portable running coach. The clothes have sensors that can monitor running-oriented metrics like cadence, ground contact time, pelvic rotation and stride length, and provide real-time feedback through headphones.

A little more advanced than the rest of the breed, Athos clothing is woven with advanced medical technology like micro-EMG sensors that detect the muscles that are working, and transfer this data to an app via Bluetooth. Apart from measuring effort, heart rate and breathing, the app also provides insights to correct your methods and avoid injury.

Baby care. The smart clothing industry is betting big on baby care, because wearables of other kinds are obviously not comfortable for babies. Owlet Smart Sock moni-

tors the baby's heart rate using pulse oximetry to make sure the baby's breathing and sleep are normal. It charges through a small base station and pairs up with a smartphone through Bluetooth.

Neopenda Smart Baby Hat has a vital signs monitor fitted inside a hat for newborns. Developed by Sona Shah and Teresa Cauvel, two biomedical engineering graduates from Columbia University, the smarthat measures temperature, heart rate, respiratory rate and blood oxygen saturation. It connects via Bluetooth to a mobile device running a custom app. The app can sync up to 24 baby hats, which can help doctors and nurses watch over all kids in a bay.

Lending a helping hand to premature babies is ARTUS, devised by Hohenstein Institute in Bönningheim. They have used smart textiles to develop an artificial uterus, which provides the sensory stimulation required for premature babies to develop. The smart textile provides the baby with acoustic stimuli like the mother's heartbeat and voice, as well as mechanical sensations like the gentle rocking experienced in the mother's body. It is currently being assessed by neonatologists.

Suits for near-field communication. Near-field communication (NFC) is a protocol for sharing data, contactless payments and so on. Smart clothing imbuing NFC technology is trending now. Samsung and Rogatis have developed an NFC smartsuit that lets the wearers do a lot of things like exchange business cards, unlock their phones, control their gadgets and more.

Barclaycard and Lyle & Scott have also designed a contactless payment jacket powered by bPay. The right cuff of this jacket hides a contactless payment chip, like the ones found in debit/credit cards. Wearers can pay for values up to £ 30 across 300,000 outlets in the UK, with just a wave of their hand.

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